Field access audit: plytix-pim-kernel v3 vs plytix-pim v1

Generated 2026-10-08 by reading source only; no runtime verification. Rows marked 'unknown' or 'unverified' in notes need a check.

One row per (entity, field). Nested fields use dotted paths. Entities = router keys in kernel api/v3/router.py, excluding product_family_models.

ACCESS COLUMNS (same vocabulary for v1 and v3)

- *_create = write | no : accepted in the create request (v1 schema / v3 input DTO), effective after hooks/handlers
- *_update = write | no : accepted in the update (PATCH) request
- *_read = exposed | hidden : present in the read output, i.e. the response (v3: GET/SEARCH output, hidden_fields, internal_fields)
- internal = no/no/hidden. read-only = no/no/exposed. read-write = write/write/exposed. Anything else is mixed (e.g. create-only).

DB OPTIONALITY

required | nullable | may-be-absent | has-default(value) | computed-not-stored | unknown (may combine with +). has-default(value) is the model default and says nothing about presence on its own. Presence rule for v1 (mongoengine 0.29.1 _delta): on update, a changed field that is falsy (not a number/bool) and equal to its default is $unset. So has-default([]) or ({}) is marked +may-be-absent (stored right after create, removed by any later save that leaves it empty); defaults of 0, False, now, enums and [[]] stay stored. Applied by rule, not checked field by field: paths written with queryset.update / $set bypass it, and explicit unset code is noted per row.

MISMATCH KIND

- none: v1 and v3 agree on create/update/response.
- moved-to-link-endpoint: v1 accepts the field in the body (id/label list), v3 does it through dedicated LINK/UNLINK endpoints. Not a real gap.
- real: v1 and v3 disagree, or a side is unknown. See 'mismatch' and 'notes'.

ENTITY ORIGIN

v3-only views (family-attributes, related-products) have no v1 model; their v1 columns describe the equivalent v1 endpoints.

YOUR COLUMNS (yellow, end of the fields tab): v3_target_create and v3_target_update (dropdown write / no), v3_target_read (dropdown exposed / hidden), and decision_notes. Same vocabulary as the v1_* and v3_* columns: internal = no/no/hidden, read-only = no/no/exposed, read-write = write/write/exposed. Left blank.

COLOUR LEGEND for db_optionality (priority: unknown > absent > nullable > computed > has-default > required)

required: always stored, never null

has-default(value): model default; stored unless it is an empty list/dict (those get +may-be-absent)

nullable: may hold null (key present)

may-be-absent (also nullable+may-be-absent): key can be missing in the DB

computed-not-stored: derived or relation, not a stored field

unknown: not determined, see notes

entity	origin	fields	real mismatches	moved to link endpoint	db nullable or absent	v3 exposed	decided (all 3 targets filled)	change needed (target differs from v3)
products	v1-native	48	12	5	25	15	48	6
product-relationships	v1-native	9	3	0	0	4	9	1
related-products	v3-only view	0	0	0	0	0	0	0
relationships	v1-native	0	0	0	0	0	0	0
assets	v1-native	40	8	1	17	18	40	15
asset-categories	v1-native	18	6	0	6	9	18	5
asset-lists	v1-native	19	1	0	7	6	19	0
product-categories	v1-native	17	6	0	6	9	17	4
product-lists	v1-native	18	1	0	6	6	18	0
product-attribute-groups	v1-native	8	3	1	2	6	8	3
product-families	v1-native	16	2	0	5	7	16	6
family-attributes	v3-only view	6	2	0	1	3	6	0
product-attributes	v1-native	34	10	0	18	19	34	3
channels	v1-native	169	41	0	91	142	169	0
connections	v1-native	21	4	0	14	13	21	0
ecatalogs	v1-native	176	7	0	146	164	176	0
pdf-catalogs	v1-native	43	3	0	25	33	43	0
import-profiles	v1-native	39	2	0	17	30	39	0
TOTAL		**681**	**111**	**7**	**386**	**484**	**681**	**43**

entity	entity_origin	field	v1_field	db_optionality	v1_create	v1_update	v1_read	v3_create	v3_update	v3_read	mismatch_kind	mismatch	v3_target_create	v3_target_update	v3_target_read	change_needed	decision_notes
products	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
products	v1-native	account_id	account_id (not in v1 schema	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
products	v1-native	sku	sku	required	write	write	exposed	write	no	exposed	real	update: v1=write v3=no	write	write	exposed	yes	
products	v1-native	label	label	may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
products	v1-native	gtin	gtin	may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
products	v1-native	status	status	has-default(Draft)+nullabl	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
products	v1-native	attributes	attributes	has-default({})+may-be-at	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
products	v1-native	attributes.<attribute_label>	attributes.<label>	may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
products	v1-native	asset_ids	asset_ids (v1 response expo	has-default([])+may-be-at	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
products	v1-native	assets	assets	computed-not-stored	no	no	exposed	no	no	exposed (exp	none		no	no	exposed (expansion)	no	
products	v1-native	thumbnail_id	thumbnail	nullable+may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
products	v1-native	thumbnail	thumbnail	computed-not-stored	no	no	exposed	no	no	exposed (exp	none		no	no	exposed (expansion)	no	
products	v1-native	category_ids	category_ids	has-default([])+may-be-at	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
products	v1-native	categories	categories	computed-not-stored	write	no	exposed	no	no	exposed (exp	oved-to-link-endpoi	[link] create: v1=write v3=no	no	no	exposed (expansion)	no	
products	v1-native	static_list_ids	static_lists_ids (response); st	has-default([])+may-be-at	write	no	exposed	no	no	hidden	real	create: v1=write v3=no; response: v1=exp	no	no	hidden	no	
products	v1-native	static_lists	static_lists	computed-not-stored	no	no	hidden	no	no	exposed (exp	oved-to-link-endpoi	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
products	v1-native	channel_ids	channel_ids	nullable+may-be-absent	write	no	exposed	no	no	hidden	real	create: v1=write v3=no; response: v1=exp	no	no	hidden	no	
products	v1-native	channels	channel_ids	computed-not-stored	write	no	exposed	no	no	exposed (exp	oved-to-link-endpoi	[link] create: v1=write v3=no	no	no	exposed(internal)	yes	
products	v1-native	ecatalog_ids	ecatalog_ids	nullable+may-be-absent	write	no	exposed	no	no	hidden	real	create: v1=write v3=no; response: v1=exp	no	no	hidden	no	
products	v1-native	ecatalogs	ecatalog_ids	computed-not-stored	write	no	exposed	no	no	exposed (exp	oved-to-link-endpoi	[link] create: v1=write v3=no	no	no	exposed(internal)	yes	
products	v1-native	product_data_sheet_ids	product_data_sheet_ids	nullable+may-be-absent	write	no	exposed	no	no	hidden	real	create: v1=write v3=no; response: v1=exp	no	no	hidden	no	
products	v1-native	product_data_sheets	product_data_sheet_ids	computed-not-stored	write	no	exposed	no	no	exposed (exp	oved-to-link-endpoi	[link] create: v1=write v3=no	no	no	exposed(internal)	yes	
products	v1-native	parent_id	_parent_id (stored as parent_	nullable+may-be-absent	no	no	exposed	write	write	exposed	real	create: v1=no v3=write; update: v1=no v3	write	write	exposed	no	
products	v1-native	parent	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
products	v1-native	num_variations	num_variations	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
products	v1-native	product_level	product_level	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
products	v1-native	product_type	product_type	computed-not-stored	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden (hidde	no	no	hidden	no	
products	v1-native	product_family_id	product_family_id	may-be-absent	write	no	exposed	write	write	exposed	real	update: v1=no v3=write	write	write	exposed	no	
products	v1-native	product_family	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
products	v1-native	product_family_model_id	product_family_model_id	may-be-absent	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	overwritten_attributes	overwritten_attributes	has-default([])+may-be-at	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
products	v1-native	mark_as_deleted	n/a	has-default(False)	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
products	v1-native	relationships	relationships	has-default([])+may-be-at	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	product_relationships_to	relationships	has-default([])+may-be-at	no	no	exposed	no	no	exposed (exp	none		no	no	exposed (expansion)	no	
products	v1-native	product_relationships_from	relationships	has-default([])+may-be-at	no	no	exposed	no	no	exposed (exp	none		no	no	exposed (expansion)	no	
products	v1-native	created	created	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
products	v1-native	modified	modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
products	v1-native	last_context	last_context	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
products	v1-native	modified_user_audit	modified_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	modified_user_audit.type	modified_user_audit.type	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	modified_user_audit.user_id	modified_user_audit.user_id	nullable	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	modified_user_audit.user_email	modified_user_audit.user_em	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	created_user_audit	created_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	created_user_audit.type	created_user_audit.type	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	created_user_audit.user_id	created_user_audit.user_id	nullable	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	created_user_audit.user_email	created_user_audit.user_em	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
products	v1-native	_created_at	n/a	has-default(now)+may-be	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
products	v1-native	resolved_fields	resolved_fields	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-relationship_types	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-relationship_types	v1-native	account_id	n/a (not in v1 schema)	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-relationship_types	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-relationship_types	v1-native	label	label	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-relationship_types	v1-native	status	status (not in v1 schema)	has-default(ACTIVE)	no	no	hidden	no	no	exposed	real	SCOPE SEARCH TO ACTIVE	no	no	exposed(internal)	yes	
product-relationship_types	v1-native	product_relationships	relationships	has-default([])+may-be-at	no	no	exposed	no	no	exposed (exp	none		no	no	exposed (expansion)	no	
product-relationship_types	v1-native	created	created (also created_at)	has-default(now)+may-be	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-relationship_types	v1-native	created_at	created_at	has-default(now)+may-be	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
product-relationship_types	v1-native	modified	modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-relationship_types	v1-native	last_context	last_context	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-relationship_types	v1-native	modified_user_audit	modified_user_audit	may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-relationship_types	v1-native	created_user_audit	created_user_audit	may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-relationships	v1-native	from_product_id	product_id (path param; also	computed-not-stored	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-relationships	v1-native	from_product	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real		no	no	exposed (expansion)	no	
product-relationships	v1-native	to_product_id	product_id (path param; also	computed-not-stored	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-relationships	v1-native	to_product	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real		no	no	exposed (expansion)	no	
product-relationships	v1-native	product_relationship_type_id	relationship_id	required	write	no	exposed	write	no	exposed	none		write	no	exposed	no	
product-relationships	v1-native	product_relationship_type	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
product-relationships	v1-native	account_id	n/a	computed-not-stored	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-relationships	v1-native	relationship_label	relationship_label	required	no	no	exposed	no	no	exposed	none	remove from v3	no	no	hidden	yes	
product-relationships	v1-native	action	action (internal API only: ADD	computed-not-stored	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
assets	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
assets	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
assets	v1-native	assigned	assigned	has-default(False)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
assets	v1-native	content_type	content_type	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	checksum	n/a	required	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed(internal)	yes	
assets	v1-native	extension	extension	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	file_size	file_size	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	file_type	file_type	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	file_modified	file_modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	filename	filename	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
assets	v1-native	path	n/a	nullable+may-be-absent	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed(internal)	yes	
assets	v1-native	downloaded_from	n/a	nullable+may-be-absent	no	no	hidden	no	no	exposed	real	USE FOR CREATE DTO (maybe rename	write	no	exposed(internal)	yes	
assets	v1-native	n_catalogs	n_catalogs	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	n_products	n_products	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
assets	v1-native	public	public	nullable+may-be-absent	write	write	exposed	no	no	exposed	real	create: v1=write v3=no; update: v1=write	no	no	exposed(internal)	yes	
assets	v1-native	alt_text	alt_text	nullable+may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
assets	v1-native	category_ids	n/a	has-default([])+may-be-at	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
assets	v1-native	categories	categories	has-default([])+may-be-at	no	write	exposed	no	no	exposed (exp	oved-to-link-endpoi	[link] update: v1=write v3=no	no	no	exposed (expansion)	no	
assets	v1-native	status	status	has-default(ACTIVE)	no	no	exposed	no	no	exposed	none	SCOPE SEARCH TO ACTIVE	no	no	exposed(internal)	yes	
assets	v1-native	static_list_ids	n/a	has-default([])+may-be-at	no	no	hidden	no	no	hidden	none		no	no	hidden	no	

entity	entity_origin	field	v1_field	db_optionality	v1_create	v1_update	v1_read	v3_create	v3_update	v3_read	mismatch_kind	mismatch	v3_target_create	v3_target_update	v3_target_read	change_needed	decision_notes
assets	v1-native	static_lists	n/a	has-default([])+may-be-at	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
assets	v1-native	created	created	has-default(now)+may-be	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
assets	v1-native	modified	modified	has-default(now)+may-be	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
assets	v1-native	url	url	computed-not-stored	write	no	exposed	write	no	hidden	real	DONT USE FOR CREATE	no	no	exposed	yes	
assets	v1-native	content	content	computed-not-stored	write	no	hidden	write	no	hidden	none		write	no	hidden	no	
assets	v1-native	thumbnail	thumbnail	computed-not-stored	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	exposed(internal)	yes	
assets	v1-native	has_custom_thumb	has_custom_thumb	computed-not-stored	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	exposed(internal)	yes	
assets	v1-native	_thumbnail	n/a	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
assets	v1-native	_thumbnail.path	n/a	required	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
assets	v1-native	_thumbnail.modified	n/a	has-default(now)	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
assets	v1-native	created_user_audit	created_user_audit	nullable+may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	created_user_audit.type	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	created_user_audit.user_id	n/a	nullable	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	created_user_audit.user_email	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	modified_user_audit	modified_user_audit	nullable+may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	modified_user_audit.type	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	modified_user_audit.user_id	n/a	nullable	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	modified_user_audit.user_email	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
assets	v1-native	last_context	n/a	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
assets	v1-native	_created_at	n/a	has-default(now)+may-be	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-categories	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
asset-categories	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
asset-categories	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
asset-categories	v1-native	parent_id	parent_id	nullable+may-be-absent	write	write	hidden	write	write	exposed	real	response: v1=hidden v3=exposed	write	write	exposed	no	
asset-categories	v1-native	parent	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
asset-categories	v1-native	parents_ids	parents_ids	has-default([])+may-be-at	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-categories	v1-native	parents	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed (in v1 t	no	no	exposed (expansion)	no	
asset-categories	v1-native	path	path	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
asset-categories	v1-native	slug	slug	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
asset-categories	v1-native	order	order	required	no	no	exposed	no	no	exposed	none		write	write	exposed	yes	
asset-categories	v1-native	sort_children	sort_children	computed-not-stored	no	write	hidden	no	no	hidden	real	update: v1=write v3=no	no	no	exposed(internal)	yes	
asset-categories	v1-native	n_children	n_children	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
asset-categories	v1-native	created	n/a (stored as _created_at)	has-default(now)+may-be	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed	no	
asset-categories	v1-native	modified	modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
asset-categories	v1-native	created_user_audit	n/a	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-categories	v1-native	modified_user_audit	n/a	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-categories	v1-native	last_context	n/a	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-categories	v1-native	(delete)	n/a	n/a	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-lists	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
asset-lists	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
asset-lists	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
asset-lists	v1-native	type	type	computed-not-stored	write	no	exposed	write	no	exposed	none		write	no	exposed	no	
asset-lists	v1-native	_cls	n/a	required	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-lists	v1-native	query	filters	has-default([[]])	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
asset-lists	v1-native	relationship_filters	relationship_filters	unknown	write	no	exposed	no	no	hidden	real	create: v1=write v3=no; response: v1=exp	no	no	hidden	no	
asset-lists	v1-native	created	_created_at	has-default(now)+may-be	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
asset-lists	v1-native	modified	modified	has-default(now)+may-be	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
asset-lists	v1-native	created_user_audit	created_user_audit	nullable+may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	created_user_audit.type	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	created_user_audit.user_id	n/a	nullable	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	created_user_audit.user_email	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	modified_user_audit	modified_user_audit	nullable+may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	modified_user_audit.type	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	modified_user_audit.user_id	n/a	nullable	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	modified_user_audit.user_email	n/a	required	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
asset-lists	v1-native	last_context	n/a	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
asset-lists	v1-native	(assets membership)	n/a	computed-not-stored	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-categories	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-categories	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-categories	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-categories	v1-native	parent_id	parent_id	nullable+may-be-absent	no	write	hidden	write	write	exposed	real	create: v1=no v3=write; response: v1=hid	write	write	exposed	no	
product-categories	v1-native	parent	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
product-categories	v1-native	parents_ids	parents_ids	has-default([])+may-be-at	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
product-categories	v1-native	parents	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
product-categories	v1-native	path	path	required	no	no	exposed	no	no	exposed	none		no	no	exposed internal ?	yes	
product-categories	v1-native	slug	slug	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
product-categories	v1-native	order	order	required	no	no	exposed	no	no	exposed	none		write	write	exposed	yes	
product-categories	v1-native	n_children	n_children	required	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
product-categories	v1-native	created	_created_at (not in category	may-be-absent	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed	no	
product-categories	v1-native	modified	modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-categories	v1-native	last_context	last_context (not in schema)	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-categories	v1-native	created_user_audit	created_user_audit (not in ca	may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-categories	v1-native	modified_user_audit	modified_user_audit (not in c	may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-categories	v1-native	sort_children	sort_children	computed-not-stored	no	write	hidden	no	no	hidden	real	update: v1=write v3=no	no	no	hidden	no	
product-lists	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-lists	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-lists	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-lists	v1-native	type	type (derived from _cls)	required	write	no	exposed	write	no	exposed	none		write	no	exposed	no	
product-lists	v1-native	query	filters	has-default([[]])	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-lists	v1-native	relationship_filters	relationship_filters	has-default([[]])	write	write	exposed	no	no	hidden	real	create: v1=write v3=no; update: v1=write	no	no	hidden	no	
product-lists	v1-native	created	_created_at	may-be-absent	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-lists	v1-native	modified	modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-lists	v1-native	created_user_audit	created_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	created_user_audit.type	created_user_audit.type	required (when parent pre	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	created_user_audit.user_id	created_user_audit.user_id	nullable (when parent pre	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	created_user_audit.user_email	created_user_audit.user_ema	required (when parent pre	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	modified_user_audit	modified_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	modified_user_audit.type	modified_user_audit.type	required (when parent pre	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	modified_user_audit.user_id	modified_user_audit.user_id	nullable (when parent pre	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	

entity	entity_origin	field	v1_field	db_optionality	v1_create	v1_update	v1_read	v3_create	v3_update	v3_read	mismatch_kind	mismatch	v3_target_create	v3_target_update	v3_target_read	change_needed	decision_notes
product-lists	v1-native	modified_user_audit.user_email	modified_user_audit.user_em	required (when parent pre	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-lists	v1-native	last_context	last_context (not in schema)	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-lists	v1-native	_cls	(exposed as type)	required	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-attribute-groups	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-attribute-groups	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-attribute-groups	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attribute-groups	v1-native	order	has-default(0)	has-default(0)	write	write	exposed	no	write	exposed	none		write	write	exposed	yes	
product-attribute-groups	v1-native	attribute_labels	attribute_labels	has-default([])+may-be-ab	write	write	exposed	no	no	exposed	real	[link] create: v1=write v3=no; [link] update	no	no	hidden	yes	
product-attribute-groups	v1-native	attributes						no	no	no	oved-to-link-endpoi	EXPOSE AS RELATIONSHIP	no	no	exposed	yes	
product-attribute-groups	v1-native	created	_created_at (not in group sch	may-be-absent	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed	no	
product-attribute-groups	v1-native	modified	modified (not in group schem	has-default(now)	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed	no	
product-families	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-families	v1-native	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-families	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-families	v1-native	total_products	total_products	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
product-families	v1-native	total_attributes	total_attributes	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	exposed(internal)	yes	
product-families	v1-native	total_models	total_models	has-default(0)	no	no	exposed	no	no	exposed	none		no	no	hidden	yes	
product-families	v1-native	created	created / created_at (from _c	may-be-absent	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-families	v1-native	created_at	created_at	may-be-absent	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
product-families	v1-native	modified	modified	has-default(now)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-families	v1-native	created_user_audit	created_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-families	v1-native	modified_user_audit	modified_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-families	v1-native	last_context	last_context (not in schema)	nullable+may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-families	v1-native	attributes	attributes	computed-not-stored	write	no	hidden	write	no	hidden	none		no	no	hidden	yes	
product-families	v1-native	parent_attributes	parent_attributes	computed-not-stored	write	no	hidden	write	no	hidden	none		no	no	hidden	yes	
product-families	v1-native	variant_attributes	variant_attributes	computed-not-stored	write	no	hidden	write	no	hidden	none		no	no	hidden	yes	
product-families	v1-native	family_attributes	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
family-attributes	v3-only view: no v	account_id	n/a	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
family-attributes	v3-only view: no v	product_family_id	product_family_id (URL path	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
family-attributes	v3-only view: no v	product_family	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
family-attributes	v3-only view: no v	attribute_id	attributes[] (link body) / id (GE	required	write	no	exposed	write	no	exposed	none		write	no	exposed	no	
family-attributes	v3-only view: no v	attribute	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
family-attributes	v3-only view: no v	level	attributes_level (link body) / p	nullable	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	id	id	required	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-attributes	v1-native	name	name	required	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	label	label	required	write	no	exposed	write	no	exposed	none		write	no	exposed	no	
product-attributes	v1-native	description	description	nullable+may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	status	status	has-default(ACTIVE)	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed(internal)	yes	
product-attributes	v1-native	created	created / created_at	may-be-absent	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-attributes	v1-native	created_at	created_at	may-be-absent	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	modified	modified	has-default(now at save)	no	no	exposed	no	no	exposed	none		no	no	exposed	no	
product-attributes	v1-native	account_id	n/a (never in v1 response)	required	no	no	hidden	no	no	xposed(interna	none		no	no	exposed(internal)	no	
product-attributes	v1-native	group_ids	groups	has-default([])+may-be-ab	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-attributes	v1-native	groups	groups	has-default([])+may-be-ab	no	no	exposed	no	no	exposed (exp	none		no	no	exposed (expansion)	no	
product-attributes	v1-native	family_attributes	n/a	computed-not-stored	no	no	hidden	no	no	exposed (exp	real	response: v1=hidden v3=exposed	no	no	exposed (expansion)	no	
product-attributes	v1-native	product_family_ids	product_family_ids	may-be-absent	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	product_families	product_families	computed-not-stored	write	no	hidden	no	no	hidden	real	create: v1=write v3=no	no	no	hidden	no	
product-attributes	v1-native	type	type_class	computed-not-stored (sto	write	no	exposed	write	no	exposed	none		write	no	exposed	no	
product-attributes	v1-native	_cls	_cls	required	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-attributes	v1-native	last_context	last_context	may-be-absent	no	no	hidden	no	no	hidden	none		no	no	hidden	no	
product-attributes	v1-native	created_user_audit	created_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	modified_user_audit	modified_user_audit	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	created_user_audit.*	created_user_audit.*	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	modified_user_audit.*	modified_user_audit.*	may-be-absent	no	no	exposed	no	no	hidden	intentional	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	filter_type	filter_type	computed-not-stored	no	no	exposed	no	no	hidden	real	response: v1=exposed v3=hidden	no	no	hidden	no	
product-attributes	v1-native	character_limit	character_limit	nullable+may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	include_time	include_time	has-default(False)	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	options	options	has-default([])+may-be-ab	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	manual_sorting	manual_sorting	has-default(True)	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	sort_ascending	sort_ascending	nullable+has-default(True	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	restricted	restricted	has-default(False)	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	renamed_options	renamed_options	computed-not-stored	no	write	hidden	no	no	hidden	real	update: v1=write v3=no	no	no	hidden	no	
product-attributes	v1-native	formula	formula	nullable+may-be-absent	write	write	exposed	write	no	exposed	real	update: v1=write v3=no	no	no	exposed(internal)	yes	
product-attributes	v1-native	formula_str	formula_str	nullable+may-be-absent	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	formula_attribute_labels	n/a	has-default([])+may-be-ab	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed(internal)	yes	
product-attributes	v1-native	attributes	attributes	has-default([])+may-be-ab	write	write	exposed	write	write	exposed	none		write	write	exposed	no	
product-attributes	v1-native	system attributes (label,status,categories,thumbnail,channel_ids,ecata	n/a (v1 builds in memory)	computed-not-stored	no	no	hidden	no	no	exposed	real	response: v1=hidden v3=exposed	no	no	exposed	no	

product_relationship_ty

	name	label	status
1	A	a	ACTIVE
2	B	b	ACTIVE
3	C	c	DELETED

from_product

from_product_id
1
3
2

to_product

to_product_id
2
1
3

product_relationship_type

product_relationship_type_id
2
1
2

properties
{quantity: X}
{quantity: X}
{quantity: X}

product_id	sku	attributes	
1	A	a	
2	B	b	
3	C	c	

Request	body/response
/products/<id>/pr	{id: <product_realtionship_type_id>, from_product_id: <product_id>, to_product_id: <product_id>, properties: {quantity: X}} IMPROTANT: only from_product_id or to_product_id, not both
/products/<id>/pr	{ [{ "from_product_id": ... "to_product_id": ... "product_relationship_type_id": ... "properties": { "quantity":... } }] }
/products/<id>/pr	{ [{ "from_product_id": ... "to_product_id": ... "product_relationship_type_id": ... "properties": { "quantity":... } }] }

	/products?_fie	{
		[{
		"from_product_id": ...
		"from_product": {
		"sku": ...
		}
		"to_product_id": ...
		"product_relationship_type_id": ...
		"product_relationship_type": {
		"id": ...
		"name": ...
		...
		}
		"properties": {
		"quantity":...
		}
		}]
		}							

	/products?product_relationships_to.properties.quantity[gte]=2								
	/products?product_relationships_to.relationship.label=requires								
	/products?product_relationships_to.id=<id>								